

QUANTUM MECHANICS

Foundations and Applications

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1 FUNDAMENTALS

This section shall cover the fundamentals of Quantum Mechanics, starting with the vector formalism, the Schrödinger Equation and other concepts, so that one is ready to solve any problems they encounter in the Wave Mechanics part of these notes. I've chosen to take this approach as I believe that the most optimal way to study physics is to start by having a strong mathematical formalism. You will only require a knowledge of introductory Linear algebra, Integral and Differential calculus to start.

1.1 Bra Ket Notation

In Classical Mechanics, the movement of a system is dictated by its initial conditions and the forces acting on it. As long as you know all of these, you can predict any future that the system has. This is the concept of *trajectory*, it is the path that a particle traces over time, which can one obtain by solving Newton's Equation: $F = ma$. Quantum Mechanics is however, not so simple. We shall start with the concept of *the state*. The state is the entity that contains all information about the system, and by the language of Linear Algebra, we can extract the information out from this system. The state is a vector, specifically a *ket vector*, represented by $|\mathcal{S}\rangle$. The state ket lives in what is called a *Hilbert Space*, which is an abstract vector space represented by $\mathcal{H}$.

1.1.1 Kets, Bras and Inner Products

The ket is an abstract vector that lives in a vector space called the Hilbert space. A ket is represented by $|\cdot\rangle$. Being a vector, obviously it has the standard properties that a vector space possesses.

- The addition of two kets gives us another ket.

$$|\alpha\rangle + |\beta\rangle = |\gamma\rangle \quad (1.1)$$

- The product of a scalar and ket is a ket, if the constant is zero then the ket produced is the null ket.

$$c(|\alpha\rangle) = c|\alpha\rangle = |\alpha\rangle c \quad (1.2)$$

Then there is the *bra*, which is essentially a mirror of a ket.

Postulate 1.1: Dual Correspondence

For every ket $|\alpha\rangle$ in a Hilbert space $\mathcal{H}$, there exists a bra $\langle\alpha|$ in the Hilbert Dual space $\mathcal{H}_d$. The bra and ket are said to have a Dual Correspondence.

$$|\alpha\rangle \xleftrightarrow{DC} \langle\alpha|$$

Bras work the same way as kets do, both are closed over addition and scalar multiplication.

In a finite dimensional vector space of N dimensions, kets and bras can be thought of as a column

and row matrix respectively. If in some basis, the ket is represented as a column matrix, then the dual bra of the ket is the *hermitian adjoint* of the ket. A hermitian adjoint is just the transpose conjugate of a matrix.

$$|\alpha\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \end{pmatrix} \xleftrightarrow{DC} \begin{pmatrix} \alpha_1^* & \alpha_2^* & \alpha_3^* & \dots \end{pmatrix} = \langle\alpha| \quad (1.3)$$

And in general,

$$a|\alpha\rangle + b|\beta\rangle \xleftrightarrow{DC} a^*\langle\alpha| + b^*\langle\beta| \quad (1.4)$$

The *inner product* is defined by the product of a bra to the left of the ket, and it is a complex scalar. You will also see the inner product to be referred to as a scalar product, but the inner product is just a generalization of the geometric scalar product.

$$\langle\alpha|\beta\rangle = \langle\alpha||\beta\rangle$$

Note that unlike the scalar product, $\langle\alpha|\beta\rangle \neq \langle\beta|\alpha\rangle$, but instead they are conjugates of each other, i.e.

$$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^* \quad (1.5)$$

Postulate 1.2: Positive Definite Metric

The inner product of a ket $|\alpha\rangle$ and its bra dual $\langle\alpha|$ is always real and positive and is called the norm, squared. It is zero if and only if the null ket is taken.

$$||\alpha||^2 = \langle\alpha|\alpha\rangle \geq 0$$

The inner product of $\langle\alpha|$ with a sum of kets $|\beta\rangle + |\gamma\rangle$ is the sum of the inner products, i.e.

$$\langle\alpha| [|\beta\rangle + |\gamma\rangle] = \langle\alpha|\beta\rangle + \langle\alpha|\gamma\rangle \quad (1.6)$$

The same applies to an inner product of a sum of bras and a ket.

$$[\langle\alpha| + \langle\beta|] |\gamma\rangle = \langle\alpha|\gamma\rangle + \langle\beta|\gamma\rangle \quad (1.7)$$

The inner product in an N dimensional space is defined as:

$$\langle\alpha|\beta\rangle = \begin{pmatrix} \alpha_1^* & \alpha_2^* & \dots & \alpha_n^* \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} = \alpha_1^*\beta_1 + \alpha_2^*\beta_2 + \dots + \alpha_n^*\beta_n \quad (1.8)$$

1.1.2 Operators and Outer Products

An operator is a map from one vector space to another, typically represented by $\hat{A}: V_1 \rightarrow V_2$. An operator acts on kets to produce another ket, and it is analogous to a linear transformation by a matrix. In fact, for a finite dimensional space with N dimensions, any operator can be represented as an $N \times N$ matrix. It being like a matrix, is also a linear entity, and therefore:

$$\hat{A}[a|\alpha\rangle + b|\beta\rangle] = a\hat{A}|\alpha\rangle + b\hat{A}|\beta\rangle \quad (1.9)$$

Operators can be multiplied, and the multiplication is associative and distributive.

$$\hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C} = \hat{A}\hat{B}\hat{C} \quad (1.10)$$

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C} \quad (1.11)$$

However, it may or may not always be commutative. The non-commutativity of operators is so important in Quantum Mechanics that we have two quantities that measure the commutativity of two operators: *the commutator and anti-commutator*, defined as follows, respectively:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \quad (1.12)$$

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A} \quad (1.13)$$

Commutators typically follow two important identities:

Leibniz Identity

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} - \hat{B}[\hat{A}, \hat{C}] \quad (1.14)$$

Jacobi Identity

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \quad (1.15)$$

The dual correspondent of $\hat{A}|\alpha\rangle$ is $\langle\alpha|\hat{A}^\dagger$ where $\hat{A}^\dagger$ is the hermitian conjugate of $\hat{A}$. An operator such that $\hat{A} = \hat{A}^\dagger$, where both have the same domain, is called a *hermitian operator*. For any two operators $\hat{A}$ and $\hat{B}$,

$$\langle\alpha|\hat{A}|\beta\rangle = [\langle\beta|\hat{A}^\dagger|\alpha\rangle]^* \quad (1.16)$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger \quad (1.17)$$

The *outer product* of a ket and bra is defined with the ket to the left and the bra to the right, and it is an operator, since it has the property of "rotating" a ket in the direction of the ket in

the outer product.

$$[|\alpha\rangle\langle\beta|]|\gamma\rangle = |\alpha\rangle\langle\beta|\gamma\rangle \quad (1.18)$$

In an N dimensional space, the outer product is as follows:

$$|\alpha\rangle\langle\beta| = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \begin{pmatrix} \beta_1^* & \beta_2^* & \dots & \beta_n^* \end{pmatrix} = \begin{pmatrix} \alpha_1\beta_1^* & \alpha_1\beta_2^* & \dots & \alpha_1\beta_n^* \\ \alpha_2\beta_1^* & \alpha_2\beta_2^* & \dots & \alpha_2\beta_n^* \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n\beta_1^* & \alpha_n\beta_2^* & \dots & \alpha_n\beta_n^* \end{pmatrix} \quad (1.19)$$

If we have a vector $|\gamma\rangle$ that is normalized, then the outer product $|\gamma\rangle\langle\gamma|$ is called a projection operator, represented by Λ_γ .

$$\Lambda_\gamma = |\gamma\rangle\langle\gamma| \quad (1.20)$$

1.1.3 Eigenvalues, Eigenkets and Observables

The eigenvalue equation of an operator $\hat{A}$, with eigenvalue A and corresponding eigenket $|A\rangle$. Typically, we only consider normalized eigenkets i.e. $\langle A|A\rangle = 1$ since $|A\rangle$ and $c|A\rangle$ represent the same state.

$$\hat{A}|A\rangle = A|A\rangle \quad (1.21)$$

Hermitian operators have a big role to play in this: they represent an experimental *observable*, be it position, momentum or energy, all these are represented by a Hermitian operator. This comes from the fact that the eigenvalues of Hermitian operators are only real.

Reality of Eigenvalues and Orthogonality of Eigenkets of Hermitian Operators

Consider two eigenvalues A and A' of a Hermitian operator $\hat{A}$ with corresponding eigenkets $|A\rangle$ and $|A'\rangle$.

$$\hat{A}|A\rangle = A|A\rangle \quad \hat{A}|A'\rangle = A'|A'\rangle \quad (1.22)$$

Applying the bra $\langle A'|$ to left in the first equation,

$$\langle A'|\hat{A}|A\rangle = A\langle A'|A\rangle \implies \langle A|\hat{A}^\dagger|A'\rangle^* = A\langle A'|A\rangle$$

However, remember that we took $\hat{A}$ to a hermitian operator, therefore $\hat{A}^\dagger = \hat{A}$, and by the second eigenvalue equation,

$$\langle A|\hat{A}|A'\rangle^* = A\langle A'|A\rangle \implies [\langle A|A'|A'\rangle]^* = A\langle A'|A\rangle$$

$$A'^*\langle A'|A\rangle = A\langle A'|A\rangle \implies (A'^* - A)\langle A'|A\rangle = 0$$

If we state A and A' to be distinct eigenvalues, the only this equality is true is when $\langle A'|A\rangle = 0$. Therefore, **the eigenkets of distinct eigenvalues are orthogonal for a hermitian operator**. If we state that the two eigenvalues are the same, then we get that $A = A^*$ and therefore **the eigenvalues of a hermitian operator are only real**.

We see that the eigenkets of a hermitian are orthonormal, and can therefore form a *basis*. Any state can therefore be represented as a linear combination of eigenkets $|A_n\rangle$ of an observable A .

$$|\mathcal{S}\rangle = \sum_{n=1} c_n |A_n\rangle \quad (1.23)$$

Since the basis is orthonormal, we can do the following:

$$\begin{aligned} \langle A_m | \mathcal{S} \rangle &= \langle A_m | \left[\sum_{n=1} c_n |A_n\rangle \right] = \sum_{n=1} c_n \langle A_m | A_n \rangle = \sum_{n=1} c_n \delta_{mn} = c_m \\ \therefore c_n &= \langle A_n | \mathcal{S} \rangle \end{aligned} \quad (1.24)$$

Here, the symbol δ_{mn} is the *Krönecker Delta*, which has the following property:

$$\delta_{mn} = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases} \quad (1.25)$$

Therefore, we can finally write the state as:

$$|\mathcal{S}\rangle = \sum_{n=1} |A_n\rangle \langle A_n | \mathcal{S} \rangle \quad (1.26)$$

Notice that the argument of the summation is actually an outer product acting on a ket. If we factor it out and consider the sum of outer products to be an operator, we see that it is the *identity operator* $\mathbf{1}$.

$$|\mathcal{S}\rangle = \left[\sum_{n=1} |A_n\rangle \langle A_n| \right] |\mathcal{S}\rangle = \mathbf{1} |\mathcal{S}\rangle$$

Identity Operator

$$\hat{I} = \sum_{n=1} |A_n\rangle \langle A_n| = \sum_{n=1} \Lambda_{A_n} \quad (1.27)$$

This result is an incredibly powerful tool. Consider an operator $\hat{Q}$ in an N dimensional space. Using our new relation,

$$\hat{Q} = \hat{I} \hat{Q} \hat{I} = \left[\sum_{n=1} |A_n\rangle \langle A_n| \right] \hat{Q} \left[\sum_{m=1} |A_m\rangle \langle A_m| \right] = \sum_{n,m=1} \langle A_n | \hat{Q} | A_m \rangle |A_n\rangle \langle A_m| \quad (1.28)$$

Here, $\langle A_n | \hat{Q} | A_m \rangle$ is the matrix element of the n^{th} row and m^{th} column of $\hat{Q}$.

$$\hat{Q} \doteq \begin{pmatrix} \langle A_1 | \hat{Q} | A_1 \rangle & \langle A_1 | \hat{Q} | A_2 \rangle & \dots & \langle A_1 | \hat{Q} | A_N \rangle \\ \langle A_2 | \hat{Q} | A_1 \rangle & \langle A_2 | \hat{Q} | A_2 \rangle & \dots & \langle A_2 | \hat{Q} | A_N \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle A_N | \hat{Q} | A_1 \rangle & \langle A_N | \hat{Q} | A_2 \rangle & \dots & \langle A_N | \hat{Q} | A_N \rangle \end{pmatrix} \quad (1.29)$$

If we consider the matrix representation of observable A in the basis of its eigenkets, we get a diagonal matrix with diagonal elements being the eigenvalues.

$$\hat{A} = \sum_{n,m=1} |A_n\rangle\langle A_n|\hat{A}|A_m\rangle\langle A_m| = \sum_{n,m=1} A_n|A_n\rangle \underbrace{\langle A_n|A_m\rangle}_{\delta_{nm}} \langle A_m| = \sum_{n=1} A_n \Lambda_{A_n} \quad (1.30)$$

1.2 Measurement and Uncertainty

1.2.1 Measurement

Let us come back to the concept of a state. We know that it is a ket and can be written as a superposition of eigenkets of an observable A . If we measure the observable A and we record that we obtain the value of A is A_1 , then it is said that *the state ket is thrown into the eigenket of the measured variable*. Before the measurement, the state ket is in a superposition of eigenkets of observable A .

$$|\mathcal{S}\rangle = \sum_{n=1} |A_n\rangle\langle A_n|\mathcal{S}\rangle \quad (1.31)$$

When the measurement is performed, the system is thrown into one of the eigenkets, say $|A_1\rangle$ of observable A .

$$|\mathcal{S}\rangle \xrightarrow{\text{measurement of } A} |A_1\rangle \quad (1.32)$$

If one measures the observable A again, then one will only obtain the same eigenvalue A_1 again. Thus, the eigenket $|A_1\rangle$ is called a *determinate state of A_1* .

Postulate 1.3: Born Interpretation

If a state ket $|\mathcal{S}\rangle$ is in a superposition of determinate states of an observable A , then the probability of measurement of the eigenvalue A_n is given by:

$$P(A_n) = |\langle A_n|\mathcal{S}\rangle|^2 \quad (1.33)$$

And the sum of the probabilities of all the eigenvalues of A is unity, and is called the normalization condition.

$$\sum_{n=1} P(A_n) = \sum_{n=1} |\langle A_n|\mathcal{S}\rangle|^2 = 1 \quad (1.34)$$

This is for a singular system, but if we were to empirically measure the probability, we must conduct a great number of measurements of an identical system of the state ket $|\mathcal{S}\rangle$, a *pure ensemble*. Suppose there are two or more linearly independent eigenkets of an observable that share the same eigenvalue. Then the eigenkets are said to *degenerate states* of A . This causes a few problems, a measurement A_n of A can no longer distinguish between eigenkets, and it is not clear to what eigenket the state collapses to. To solve this problem, we must instead use an orthonormal basis $\{|\phi_{n,k}\rangle\}$ that spans the degenerate subspace, typically of an observable B .

$$|\mathcal{S}\rangle = \frac{1}{\sqrt{A_n}} \sum_{k=1} |\phi_{n,k}\rangle\langle\phi_{n,k}|\mathcal{S}\rangle \quad (1.35)$$

Let us now talk about the expectation value.

Expectation Value

In statistics, the expectation value or the mean is defined as follows:

$$\langle A \rangle := \sum_{n=1} A_n P(A_n) \quad (1.36)$$

From equation (1.33) we can substitute the postulate,

$$\langle A \rangle = \sum_{n=1} A_n |\langle A_n | \mathcal{S} \rangle|^2 = \sum_{n=1} \langle \mathcal{S} | \underbrace{A_n | A_n \rangle}_{\hat{A} | A_n \rangle} \langle A_n | \mathcal{S} \rangle = \sum_{n=1} \langle \mathcal{S} | \hat{A} | A_n \rangle \langle A_n | \mathcal{S} \rangle \quad (1.37)$$

$$\langle A \rangle = \langle \mathcal{S} | \hat{A} \left[\underbrace{\sum_{n=1} |A_n \rangle \langle A_n|}_{\hat{I}} \right] | \mathcal{S} \rangle = \langle \mathcal{S} | \hat{A} | \mathcal{S} \rangle \quad (1.38)$$

1.2.2 Compatibility of Observables

Two observables A and B are said to be *compatible* when they commute, i.e.

$$[\hat{A}, \hat{B}] = 0 \iff \hat{A}\hat{B} = \hat{B}\hat{A} \quad (1.39)$$

They are *incompatible* if they do not commute, i.e.

$$[\hat{A}, \hat{B}] \neq 0 \quad (1.40)$$

Let us first consider the case of compatible observables in the space spanned by the eigenkets of A . If the eigenvalues of A are non-degenerate, then we can prove that the matrix elements of $\hat{B}$ in the basis of eigenkets of A are diagonal.

$$\langle A_n | [\hat{A}, \hat{B}] | A_m \rangle = \langle A_n | \hat{A}\hat{B} | A_m \rangle - \langle A_n | \hat{B}\hat{A} | A_m \rangle = (A_n - A_m) \langle A_n | \hat{B} | A_m \rangle = 0 \quad (1.41)$$

$\langle A_n | \hat{B} | A_m \rangle$ will vanish unless $A_n = A_m$, hence we can write the matrix elements of $\hat{B}$ as follows:

$$\langle A_n | \hat{B} | A_m \rangle = \delta_{nm} \langle A_n | \hat{B} | A_n \rangle \quad (1.42)$$

$$\therefore \hat{B} = \sum_{n=1} |A_n \rangle \langle A_n | \hat{B} | A_n \rangle \langle A_n | \quad (1.43)$$

Suppose $\hat{B}$ acts on an eigenket of A , say A_m ,

$$\hat{B} | A_m \rangle = \left[\sum_{n=1} |A_n \rangle \langle A_n | \hat{B} | A_n \rangle \langle A_n | \right] | A_m \rangle = \left[\langle A_m | \hat{B} | A_m \rangle \right] | A_m \rangle \quad (1.44)$$

Therefore we see that A and B share eigenkets but with different eigenvalues. These are called *simultaneous eigenkets* and are represented by $|A_n, B_n \rangle$. This holds true for degenerate states.

A simultaneous eigenket $|A_n, B_n\rangle$ has the following property:

$$\hat{A}|A_n, B_n\rangle = A_n|A_n, B_n\rangle \quad (1.45)$$

$$\hat{B}|A_n, B_n\rangle = B_n|A_n, B_n\rangle \quad (1.46)$$

If we have a set of compatible observables $A, B, C, \dots$, then the simultaneous eigenkets of $A_n, B_n, C_n, \dots$ is uniquely specified and denoted by a collective index $K_n = (A_n, B_n, C_n, \dots)$.

$$|K_n\rangle = |A_n, B_n, C_n, \dots\rangle \implies \langle K_n | K_m \rangle = \delta_{nm} \implies \sum_{n=1} \Lambda_{K_n} = \hat{I} \quad (1.47)$$

1.2.3 Incompatibility of Observables

Two observables are said to be *incompatible* if they do not commute, i.e.

$$[\hat{A}, \hat{B}] \neq 0 \iff \hat{A}\hat{B} \neq \hat{B}\hat{A} \quad (1.48)$$

This implies that they do not have simultaneous eigenkets $|A_n, B_n\rangle$. However, there can exist a subspace of the Hilbert ket space such that A and B enjoy simultaneous eigenkets with the following equation being true.

$$\hat{A}\hat{B}|A_n, B_n\rangle = \hat{B}\hat{A}|A_n, B_n\rangle \quad (1.49)$$

Let us discuss a measurement experiment to better understand incompatible observables.

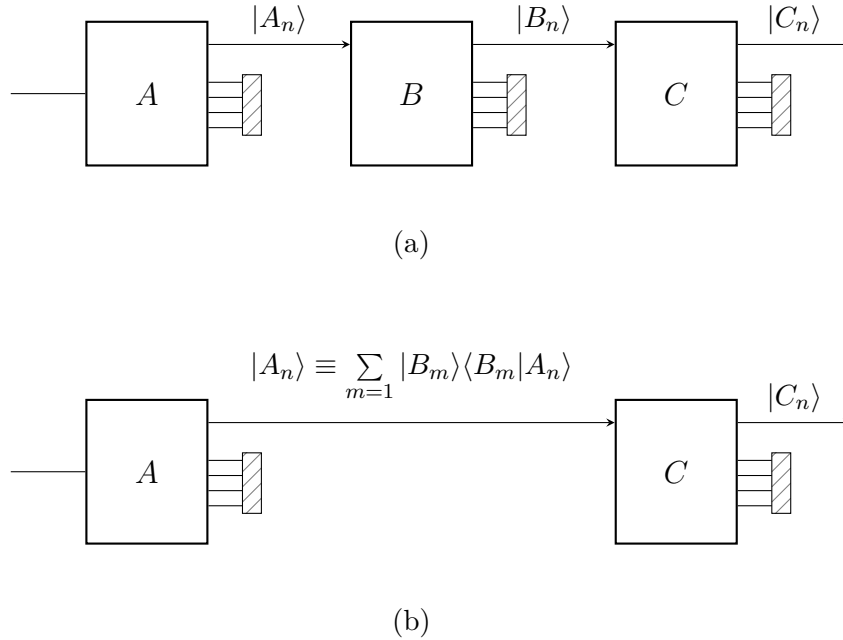


Fig 1.1: Sequential selective measurements

Consider a setup (Fig 1.1:(a)) where a state $|A_n\rangle = |A'\rangle$ is measured selectively, while all other measured states are discarded by a filter. The same is repeated for $|B_n\rangle = |B'\rangle$ and $|C_n\rangle = |C'\rangle$. The probability of a measurement of $|C'\rangle$ in the final filter would be a product of the probability that we get $|B'\rangle$ in the second filter in the eigenket basis of A and the probability that we get $|C'\rangle$ in the eigenket basis of B i.e. $|\langle C'|B'\rangle|^2 \cdot |\langle B'|A'\rangle|^2$. We repeat this procedure for all possible values of B , so we get the following sum:

$$\sum_{n=1} |\langle C'|B_n\rangle|^2 |\langle B_n|A'\rangle|^2 = \sum_{n=1} \langle C'|B_n\rangle \langle B_n|A'\rangle \langle A'|B_n\rangle \langle B_n|C'\rangle \quad (1.50)$$

If we consider a second setup (Fig 1.1(b)) where the second filter for B is inactive, the probability of measurement of C' in the final filter would be:

$$|\langle C'|A'\rangle|^2 = \left| \sum_{n=1} \langle C'|B_n\rangle \langle B_n|A'\rangle \right|^2 = \sum_{n,m=1} \langle C'|B_n\rangle \langle B_n|A'\rangle \langle A'|B_m\rangle \langle B_m|C'\rangle \quad (1.51)$$

You see that both equations are not the same! This implies that the probability indeed depends on whether a measurement of B has been taken, and that the probability are not independent of each other, unlike compatible observables. The two equations are equal if $[\hat{A}, \hat{B}] = 0$ or $[\hat{B}, \hat{C}] = 0$.

1.2.4 Uncertainty Relation

Given an observable A , we define the mean deviation operator as follows:

$$\Delta A = \hat{A} - \langle A \rangle \quad (1.52)$$

In statistics, the variance is defined as follows:

$$v = \sigma^2 = \sum_{n=1} (A_n - \langle A \rangle)^2 = \langle A^2 \rangle - \langle A \rangle^2 \quad (1.53)$$

From the new operator we just defined, we can write:

$$\sigma_A^2 = \langle (\Delta A)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2 \quad (1.54)$$

The variance, or as some books call it – dispersion – measures how much the value of A deviates from the mean value. The square root of the variance is called the uncertainty or standard deviation.

Uncertainty Principle

We consider the variance of two observables A and B .

$$\sigma_A^2 = \langle (\Delta A)^2 \rangle \quad (1.55)$$

$$\sigma_B^2 = \langle (\Delta B)^2 \rangle \quad (1.56)$$

By the Cauchy–Schwarz Inequality,

$$\sigma_A^2 \sigma_B^2 = \langle (\Delta A)^2 (\Delta B)^2 \rangle \geq |\langle \Delta A \Delta B \rangle|^2 \quad (1.57)$$

And note that we can write $\Delta A \Delta B$ as follows:

$$\Delta A \Delta B = \frac{1}{2} [\hat{A}, \hat{B}] + \frac{1}{2} \{\hat{A}, \hat{B}\} \quad (1.58)$$

$$\Rightarrow |\langle \Delta A \Delta B \rangle|^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \quad (1.59)$$

$$\therefore \sigma_A^2 \sigma_B^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 \quad (1.60)$$

This is the uncertainty principle. If we consider the anti commutator term too, we get a stronger relation.

1.2.5 Unitary Operators and Change of Basis

An operator whose inverse is the same as its hermitian is called a *unitary operator*, and it follows the following identity:

$$\hat{U} \hat{U}^\dagger = \hat{U}^\dagger \hat{U} = \hat{I} \quad (1.61)$$

Now we consider an operator $\hat{U} = \sum_{n=1} |A_n\rangle \langle B_n|$ where A_n and B_n are eigenkets of observables A and B . We see that this operator is unitary.

$$\hat{U}^\dagger = \sum_{n=1} (|B_n\rangle \langle A_n|)^\dagger = \sum_{n=1} |A_n\rangle \langle B_n| \quad (1.62)$$

$$\hat{U}^\dagger \hat{U} = \sum_{n,m=1} \langle B_n | B_m \rangle \langle A_m | A_n \rangle = \sum_{n,m=1} \delta_{nm} \delta_{nm} \hat{I} = \hat{I} \quad (1.63)$$

Applying this operator to an eigenket of B , say $|B_m\rangle$:

$$\hat{U} |B_m\rangle = \sum_{n=1} |A_n\rangle \langle B_n | B_m \rangle = |A_m\rangle \quad (1.64)$$

Similarly, operating $\hat{U}^\dagger$ to $|A_m\rangle$,

$$\hat{U}^\dagger |A_m\rangle = \sum_{n=1} |B_n\rangle \langle A_n | A_m \rangle = |B_m\rangle \quad (1.65)$$

We see that the unitary operator transforms eigenkets of B to A and its hermitian, vice-versa. For some arbitrary ket $|\alpha\rangle$ in the basis of eigenkets of A , we multiply $\langle B_m|$ to the left:

$$\langle B_m|\alpha\rangle = \sum_{n=1} \langle B_m|A_n\rangle \langle A_n|\alpha\rangle = \sum_{n=1} \langle A_m|\hat{U}^\dagger|A_n\rangle \langle A_n|\alpha\rangle \quad (1.66)$$

Therefore, we get the elements of $|\alpha\rangle$ in the basis of eigenkets of B by just operating $\hat{U}^\dagger$.

$$\langle B_n|\alpha\rangle = \langle A_n|\hat{U}^\dagger|\alpha\rangle \iff |\alpha\rangle = \sum_{n=1} |B_n\rangle \langle A_n|\hat{U}^\dagger|\alpha\rangle \quad (1.67)$$

Similarly, the matrix elements for an operator are given by:

$$\langle B_m|\hat{Q}|B_n\rangle = \sum_{i,j=1} \langle B_m|A_i\rangle \langle A_i|\hat{Q}|A_j\rangle \langle A_j|B_n\rangle \quad (1.68)$$

$$= \sum_{i,j=1} \langle A_m|\hat{U}^\dagger|A_i\rangle \langle A_i|\hat{Q}|A_j\rangle \langle A_j|\hat{U}|A_n\rangle \quad (1.69)$$

$$\therefore \hat{Q}_B = \hat{U}^\dagger \hat{Q}_A \hat{U} \quad (1.70)$$

Unitary Equivalent Observables

If we instead consider an observable A , then its unitary transformation $\hat{A}_B = \hat{U}^\dagger \hat{A} \hat{U}$ has the same eigenvalues, but the eigenkets are that of the B .

$$\hat{A}|A_n\rangle = A_n|A_n\rangle \implies \hat{U} \hat{A} \hat{U}^\dagger \hat{U}|A_n\rangle = A_n \hat{U}|A_n\rangle \quad (1.71)$$

$$(\hat{U} \hat{A} \hat{U}^\dagger)|B_n\rangle = A_n|B_n\rangle \quad (1.72)$$

Then we call $\hat{A}$ and $\hat{U} \hat{A} \hat{U}^\dagger$ *unitary equivalent observables*. It means that they have identical spectra of eigenvalues. In some cases, we may see that $\hat{B}$ and $\hat{U} \hat{A} \hat{U}^\dagger$ are in fact the same.

Unitary operators can also be used diagonalize other operators. For example, if we know $\hat{B}$ in the basis of eigenkets of A , we can find the eigenvalues by diagonalizing $\hat{B}_A$.

$$\hat{B}|B_n\rangle = B_n|B_n\rangle \implies \langle A_m|\hat{B}|B_n\rangle = B_n \langle A_m|B_n\rangle \quad (1.73)$$

$$\sum_{n=1} \langle A_m|\hat{B}|A_n\rangle \langle A_n|B_n\rangle = B_n \langle A_m|B_n\rangle \quad (1.74)$$

In matrix form, we can write it as:

$$\begin{pmatrix} B_{11} & B_{12} & \dots \\ B_{21} & B_{22} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} C_{1m} \\ C_{2m} \\ \vdots \end{pmatrix} = B_m \begin{pmatrix} C_{1m} \\ C_{2m} \\ \vdots \end{pmatrix} \quad (1.75)$$

$$B_{ij} = \langle A_i|\hat{B}|A_j\rangle \quad C_{nm} = \langle A_n|B_m\rangle \quad (1.76)$$

The matrix formed by the elements C_{nm} is the unitary operator that diagonalizes $\hat{B}_A$ into $\hat{B}$.

1.3 Position and Momentum

1.3.1 Continuous Spectra

Until now, we only discussed observables with a discrete spectra. However, we have observables like position and momentum that take continuous values. To analyze these observables, we just swap the discrete sum with a continuous integral.

For some observable $\hat{e}$ with the following continuous eigenvalue equation, we generalize the properties of discrete spectra.

$$\hat{e}|e\rangle = e|e\rangle \quad (1.77)$$

$$\text{Orthonormality: } \langle A_n | A_m \rangle = \delta_{nm} \longrightarrow \langle e | e' \rangle = \delta(e - e') \quad (1.78)$$

$$\text{Identity Operator: } \sum_{n=1} |A_n\rangle \langle A_n| = \hat{I} \longrightarrow \int |e\rangle \langle e| de = \hat{I} \quad (1.79)$$

$$\text{Born Rule: } \sum_{n=1} |\langle A_n | \alpha \rangle|^2 = 1 \longrightarrow \int |\langle e | \alpha \rangle|^2 dz = 1 \quad (1.80)$$

$$\text{Ket Expansion: } |\alpha\rangle = \sum_{n=1} |A_n\rangle \langle A_n | \alpha \rangle \longrightarrow |\alpha\rangle = \int |e\rangle \langle e | \alpha \rangle de \quad (1.81)$$

$$\text{Matrix Expansion: } \langle A_m | \hat{A} | A_n \rangle = A_n \delta_{mn} \longrightarrow \langle e_{z'} | \hat{e} | e \rangle = e \delta(e - e') \quad (1.82)$$

Here the symbol $\delta(x)$ stands for the Dirac delta function, which has the two following properties:

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad (1.83)$$

$$\text{For } \varepsilon > 0, \int_{-\varepsilon}^{+\varepsilon} \delta(x) dx = 1 \quad (1.84)$$

1.3.2 Position

Consider the eigenvalue equation for position in one dimension:

$$\hat{x}|x\rangle = x|x\rangle \quad (1.85)$$

If we take a measurement of the position, say we obtain x' , then the system collapses to the corresponding eigenket $|x'\rangle$. Of course, this is a very ideal case, real detector will record a measurement between $x' \pm \varepsilon$.

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |x\rangle \langle x | \mathcal{S} \rangle dx \xrightarrow{\text{measurement of } x} \int_{x'-\varepsilon}^{x'+\varepsilon} |x\rangle \langle x | \mathcal{S} \rangle dx \quad (1.86)$$

And by the Born Probability Interpretation, the probability that we obtain a measurement between x_1 and x_2 is:

$$P(x_1 < x < x_2) = \int_{x_1}^{x_2} |\langle x|\mathcal{S}\rangle|^2 dx \quad (1.87)$$

Extending this into three dimensions, we can write the state ket as follows:

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r \quad (1.88)$$

where $|\mathbf{r}\rangle$ is the simultaneous eigenket of observables x, y and z .

$$|\mathbf{r}\rangle = |x, y, z\rangle \quad (1.89)$$

This implies that the observables x, y, z are all compatible with each other.

1.3.3 Translation

We introduce an operator that changes a position eigenket $|\mathbf{r}\rangle$ into $|\mathbf{r} + d\mathbf{r}\rangle$. We can say the ket is being translated by an infinitesimal amount $d\mathbf{r}$ by the *translation operator*.

$$\mathcal{T}(d\mathbf{r})|\mathbf{r}\rangle = |\mathbf{r} + d\mathbf{r}\rangle \quad (1.90)$$

Let us further analyze this by translating an arbitrary state ket.

$$\mathcal{T}(d\mathbf{r})|\mathcal{S}\rangle = \mathcal{T}(d\mathbf{r}) \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r = \int_{-\infty}^{+\infty} |\mathbf{r} + d\mathbf{r}\rangle \langle \mathbf{r}|\mathcal{S}\rangle d^3r = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r} - d\mathbf{r}|\mathcal{S}\rangle d^3r \quad (1.91)$$

The last equality is true because we are integrating over all space, and the substitution $\mathbf{r} \rightarrow \mathbf{r} - d\mathbf{r}$ will not change the integral.

We demand that the translation operator must have the following properties:

- The operator must be unitary by the fact that the probability must be conserved upon translation.

$$\langle \mathcal{S}|\mathcal{S}\rangle = \langle \mathcal{S}|\mathcal{T}^\dagger(d\mathbf{r})\mathcal{T}(d\mathbf{r})|\mathcal{S}\rangle \iff \mathcal{T}^\dagger(d\mathbf{r})\mathcal{T}(d\mathbf{r}) = \hat{I} \quad (1.92)$$

- The net result of two translations by $d\mathbf{r}$ and $d\mathbf{r}'$ should be their sum.

$$\mathcal{T}(d\mathbf{r})\mathcal{T}(d\mathbf{r}') = \mathcal{T}(d\mathbf{r} + d\mathbf{r}') \quad (1.93)$$

- The inverse of a translation should be the same as the translation in the opposite direction.

$$\mathcal{T}(-d\mathbf{r}) = \mathcal{T}^{-1}(d\mathbf{r}) = \mathcal{T}^\dagger(d\mathbf{r}) \quad (1.94)$$

- The translation such that $d\mathbf{r} \rightarrow 0$ must be an identity.

$$\lim_{d\mathbf{r} \rightarrow 0} \mathcal{T}(d\mathbf{r}) = \hat{I} \quad (1.95)$$

All these properties are satisfied if we demand that the translation operator and the identity have a difference in the first order, i.e. $(\hat{\mathbf{K}} = (\hat{K}_x, \hat{K}_y, \hat{K}_z))$ is a hermitian operator)

$$\mathcal{T}(d\mathbf{r}) = \hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r} \quad (1.96)$$

Checking the first property:

$$\mathcal{T}(d\mathbf{r})\mathcal{T}^\dagger(d\mathbf{r}) = (\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})(\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})^\dagger \quad (1.97)$$

$$= (\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})(\hat{I} + i\hat{\mathbf{K}} \cdot d\mathbf{r}) \quad (1.98)$$

$$= \hat{I} + i(0) \cdot d\mathbf{r} + \hat{\mathbf{K}}^2 [d\mathbf{r}]^2 \approx \hat{I} \quad (1.99)$$

Checking the second property:

$$\mathcal{T}(d\mathbf{r})\mathcal{T}(d\mathbf{r}') = (\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r})(\hat{I} - i\hat{\mathbf{K}} \cdot d\mathbf{r}') \quad (1.100)$$

$$= \hat{I} - i\hat{\mathbf{K}}(d\mathbf{r} + d\mathbf{r}') + \hat{\mathbf{K}}^2 \cdot d\mathbf{r} \cdot d\mathbf{r}' \quad (1.101)$$

$$\approx \hat{I} - i\hat{\mathbf{K}}(d\mathbf{r} + d\mathbf{r}') = \mathcal{T}(d\mathbf{r} + d\mathbf{r}') \quad (1.102)$$

And the third and fourth properties are also satisfied. Now, consider the following:

$$\hat{\mathbf{r}}\mathcal{T}(d\mathbf{r})|\mathbf{r}\rangle = \hat{\mathbf{r}}|\mathbf{r} + d\mathbf{r}\rangle = (\mathbf{r} + d\mathbf{r})|\mathbf{r} + d\mathbf{r}\rangle \quad (1.103)$$

$$\mathcal{T}(d\mathbf{r})\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}\mathcal{T}(d\mathbf{r})|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r} + d\mathbf{r}\rangle \quad (1.104)$$

From these two, we see that:

$$[\hat{\mathbf{r}}, \mathcal{T}(d\mathbf{r})]|\mathbf{r}\rangle = d\mathbf{r}|\mathbf{r} + d\mathbf{r}\rangle \approx d\mathbf{r}|\mathbf{r}\rangle \quad (1.105)$$

$$\therefore [\hat{\mathbf{r}}, \mathcal{T}(d\mathbf{r})] = d\mathbf{r} \quad (1.106)$$

We can generate a relation between $\hat{\mathbf{r}}$ and $\hat{\mathbf{K}}$ using this relation.

$$-i\hat{\mathbf{r}}\hat{\mathbf{K}} \cdot d\mathbf{r} + i\hat{\mathbf{K}} \cdot d\mathbf{r} \cdot \hat{\mathbf{r}} = d\mathbf{r} \quad (1.107)$$

If we check for $\hat{\mathbf{r}}$ and $\hat{\mathbf{K}}$ in each direction, we get the following relation:

$$[\hat{r}_i, \hat{K}_j] = i\delta_{ij} \quad (1.108)$$

where $i, j = x, y, z$ and $\hat{r}_x = \hat{x}$, $\hat{r}_y = \hat{y}$ and $\hat{r}_z = \hat{z}$.

1.3.4 Momentum

If we realize that momentum, classically, can be thought of a generator of translation. It just so happens that $\hat{\mathbf{K}}$ term in the translation operator is the wave number, and by the De-Broglie relation,

$$k = \frac{p}{\hbar} \implies \hat{\mathbf{K}} = \frac{1}{\hbar}\hat{\mathbf{p}} \quad (1.109)$$

Therefore, the translation operator can be written as follows:

$$\mathcal{T}(d\mathbf{r}) = \hat{I} - i\hat{\mathbf{p}} \cdot d\mathbf{r}/\hbar \quad (1.110)$$

And we obtain the following commutator relation:

Canonical Commutation Relation

$$[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij} \quad (1.111)$$

This is called canonical commutation relation of quantum mechanics. From the uncertainty principle, we can now obtain the famous Heisenberg Uncertainty Principle between position and momentum.

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2} \quad (1.112)$$

Consider a finite translation Δx in the $\mathbf{x}$ direction.

$$\mathcal{T}(\Delta x \cdot \mathbf{x})|x\rangle = |\mathbf{r} + \Delta x \cdot \mathbf{x}\rangle \quad (1.113)$$

We can achieve this N by repeated applications of the translation operator by some $d\mathbf{r} = \Delta x \cdot \mathbf{x}/N$ and letting $N \rightarrow \infty$.

$$\mathcal{T}(\Delta x \cdot \mathbf{x}) = \lim_{N \rightarrow \infty} \mathcal{T}^N \left(\frac{\Delta x}{N} \mathbf{x} \right) = \lim_{N \rightarrow \infty} \left(1 - \frac{i\hat{p}_x \Delta x}{N\hbar} \right)^N = e^{i\hat{p}_x \Delta x/\hbar} \quad (1.114)$$

By the use of a Taylor series expansion,

$$\mathcal{T}(\Delta x \cdot \mathbf{x}) = 1 - \frac{i\hat{p}_x \Delta x}{\hbar} - \frac{\hat{p}_x^2 (\Delta x)^2}{2\hbar^2} + \dots \quad (1.115)$$

Similarly, in the $\mathbf{y}$ direction we have,

$$\mathcal{T}(\Delta y \cdot \mathbf{y}) = 1 - \frac{i\hat{p}_y \Delta y}{\hbar} - \frac{\hat{p}_y^2 (\Delta y)^2}{2\hbar^2} + \dots \quad (1.116)$$

Now, we consider the commutator of these two operators

$$[\mathcal{T}(\Delta x \cdot \mathbf{x}), \mathcal{T}(\Delta y \cdot \mathbf{y})] = \mathcal{T}(\Delta x \cdot \mathbf{x})\mathcal{T}(\Delta y \cdot \mathbf{y}) - \mathcal{T}(\Delta y \cdot \mathbf{y})\mathcal{T}(\Delta x \cdot \mathbf{x}) \quad (1.117)$$

$$= \mathcal{T}(\Delta x \cdot \mathbf{x} + \Delta y \cdot \mathbf{y}) - \mathcal{T}(\Delta y \cdot \mathbf{y} + \Delta x \cdot \mathbf{x}) = 0 \quad (1.118)$$

And if we find the commutator in the series expansion form, we obtain that:

$$\frac{(\Delta x)(\Delta y)}{\hbar^2} [\hat{p}_x, \hat{p}_y] = 0 \implies [\hat{p}_x, \hat{p}_y] = 0 \quad (1.119)$$

And more generally we have:

$$[\hat{p}_i, \hat{p}_j] = 0 \quad (1.120)$$

Therefore, we have a simultaneous eigenket $|\mathbf{p}\rangle$ of p_x, p_y and p_z .

$$|\mathbf{p}\rangle = |p_x, p_y, p_z\rangle \quad (1.121)$$

And if we work out the effect of the translation operator on $|\mathbf{p}\rangle$,

$$\mathcal{T}(d\mathbf{r})|\mathbf{p}\rangle = \left(1 - \frac{i\hat{\mathbf{p}} \cdot d\mathbf{r}}{\hbar}\right) |\mathbf{p}\rangle = \left(1 - \frac{i\mathbf{p} \cdot d\mathbf{r}}{\hbar}\right) |\mathbf{p}\rangle \quad (1.122)$$

$|\mathbf{p}\rangle$ is therefore an eigenket of the translation operator, and the operators commute.

$$[\mathcal{T}(d\mathbf{r}), \hat{\mathbf{p}}] = 0 \quad (1.123)$$

1.3.5 Wave Functions

We finally arrive at a study of wave functions in the position and momentum basis. We know that any state is normalized as $\langle x'|x\rangle = \delta(x' - x)$, which leads to the following ket expansion:

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.124)$$

and we can treat $|\langle x|\mathcal{S}\rangle|^2 dx$ to be the probability of measuring x in an infinitesimal interval of dx . We interpret that inner product $\langle x|\mathcal{S}\rangle$ is the position space wave function $\psi(x)$.

$$\psi_{\mathcal{S}}(x) = \langle x|\mathcal{S}\rangle \quad (1.125)$$

The inner product between two arbitrary vectors $|\alpha\rangle$ and $\langle\beta|$ can be written as:

$$\langle\beta|\alpha\rangle = \int_{-\infty}^{+\infty} \langle\beta|x\rangle \langle x|\alpha\rangle dx = \int_{-\infty}^{+\infty} \psi_{\beta}^*(x) \psi_{\alpha}(x) dx \quad (1.126)$$

If we have some state as a superposition of eigenkets of observable A , we can write the following:

$$|\mathcal{S}\rangle = \sum_{n=1} |A_n\rangle \langle A_n|\mathcal{S}\rangle \implies \psi_{\mathcal{S}}(x) = \sum_{n=1} c_n u_n(x) \quad (1.127)$$

where $c_n = \langle A_n|\mathcal{S}\rangle$ and $u_n(x) = \langle x|A_n\rangle$ is the eigenfunction of operator $\hat{A}$ with eigenvalue A_n .

$$\hat{A}u_n(x) = A_n u_n(x) \quad (1.128)$$

The expectation value of an observable A can be written as follows:

$$\langle A \rangle = \langle \mathcal{S}|\hat{A}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \langle \mathcal{S}|x\rangle \langle x|\hat{A}|x'\rangle \langle x'|\mathcal{S}\rangle dx dx' \quad (1.129)$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi_{\mathcal{S}}^*(x) \langle x|\hat{A}|x'\rangle \psi_{\mathcal{S}}(x') dx dx' \quad (1.130)$$

If $\hat{A} = g(x)$, then we can do the following:

$$\langle A \rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi_{\mathcal{S}}^*(x) g(x) \underbrace{\langle x|x' \rangle}_{\delta(x-x')} \psi_{\mathcal{S}}(x) dx dx' \quad (1.131)$$

$$= \int_{-\infty}^{+\infty} \psi_{\mathcal{S}}^*(x) g(x) \psi_{\mathcal{S}}(x) dx \quad (1.132)$$

We can say the same about the wave function in momentum space.

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |p\rangle \langle p|\mathcal{S}\rangle dp \quad (1.133)$$

$$\phi_{\mathcal{S}}(p) = \langle p|\mathcal{S}\rangle = \sum_{n=1} c_n v_n(p) \quad (1.134)$$

$$\text{where } c_n = \langle p|A_n\rangle, \hat{A}v_n(p) = A_nv_n(p) \quad (1.135)$$

$$\langle A \rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \phi_{\mathcal{S}}^*(p) \langle p|\hat{A}|p'\rangle \phi_{\mathcal{S}}(p') dp dp' = \int_{-\infty}^{+\infty} \underbrace{\phi_{\mathcal{S}}^*(p) h(p) \phi_{\mathcal{S}}(p)}_{\hat{A}=h(p)} dp \quad (1.136)$$

1.3.6 Momentum and Position Operators

First, we may examine how the momentum operators look in the position basis. Consider the definition of momentum as a generator of translations, hence:

$$\left(1 - \frac{i\hat{p}\Delta x}{\hbar}\right) |\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \mathcal{T}(\Delta x) |x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.137)$$

$$= \int_{-\infty}^{+\infty} |x + \Delta x\rangle \langle x|\mathcal{S}\rangle dx \quad (1.138)$$

$$= \int_{-\infty}^{+\infty} |x\rangle \langle x - \Delta x|\mathcal{S}\rangle dx \quad (1.139)$$

$$= \int_{-\infty}^{+\infty} |x\rangle (\langle x|\mathcal{S}\rangle - \Delta x \partial_x \langle x|\mathcal{S}\rangle) dx \quad (1.140)$$

We get the last line by applying a Taylor series expanding and ignoring higher order terms. Now, by comparing the left and right hand sides, we obtain:

$$\hat{p}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} (-i\hbar \partial_x \langle x|\mathcal{S}\rangle) |x\rangle dx \quad (1.141)$$

By orthogonality, it implies the following:

$$\langle x|\hat{p}|\mathcal{S}\rangle = -i\hbar \partial_x \langle x|\mathcal{S}\rangle \quad (1.142)$$

And therefore, the momentum operator in the position basis is given by:

$$\therefore \langle x|\hat{p} = -i\hbar \partial_x \langle x| \quad (1.143)$$

Since we have the momentum operator purely as function of x , its expectation value is given by:

$$\langle p \rangle = \int_{-\infty}^{+\infty} \psi^*(x) [-i\hbar \partial_x \psi(x)] dx \quad (1.144)$$

Consider the momentum eigenvalue equation:

$$\hat{p}|p\rangle = p|p\rangle \implies \langle x|\hat{p}|p\rangle = p\langle x|p\rangle \quad (1.145)$$

And from earlier, we know that $\langle x|\hat{p}|\mathcal{S}\rangle = -i\hbar\partial_x\langle x|\mathcal{S}\rangle$, then we get the following differential equation:

$$\partial_x\langle x|p\rangle = \frac{ip}{\hbar}\langle x|p\rangle \quad (1.146)$$

The solution to this differential equation is $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar}$ after normalization.

Now consider the following:

$$\phi(p) = \langle p|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \langle p|x\rangle\langle x|\mathcal{S}\rangle dx \quad (1.147)$$

$$\therefore \phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-ipx/\hbar} \psi(x) dx \quad (1.148)$$

Similarly,

$$\therefore \psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{ipx/\hbar} \phi(p) dp \quad (1.149)$$

This way, we transform between the position and momentum wave function, and therefore the function $\langle x|p\rangle$ is called the transformation function of x and p .

Now, to find the position operator in momentum space, consider the following:

$$\langle p|\hat{x}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} \langle p|\hat{x}|x\rangle\langle x|\mathcal{S}\rangle dx = \int_{-\infty}^{+\infty} x\langle p|x\rangle\langle x|\mathcal{S}\rangle dx \quad (1.150)$$

Since we know that the form of $\langle x|p\rangle$ and that $\langle x|\mathcal{S}\rangle = \psi(x)$,

$$\langle p|\hat{x}|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} x e^{-ipx/\hbar} \psi(x) dx = \int_{-\infty}^{+\infty} i\hbar\partial_p(e^{-ipx/\hbar})\psi(x) dx \quad (1.151)$$

We can pull this derivative out as it is in terms of p , not x , which gives us:

$$\langle p|\hat{x}|\mathcal{S}\rangle = i\hbar\partial_p \underbrace{\int_{-\infty}^{+\infty} e^{-ipx/\hbar} \psi(x) dx}_{\phi(p)} = i\hbar\partial_p\langle p|\mathcal{S}\rangle \quad (1.152)$$

And therefore, we obtain the position operator in the momentum basis to be the following:

$$\therefore \langle p|\hat{x} = i\hbar\partial_p\langle p| \quad (1.153)$$

1.3.7 Generalization to Three Dimensions

The eigenkets of position and momentum satisfy:

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle \quad \hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle \quad (1.154)$$

They obey the normalization conditions

$$\langle \mathbf{r} | \mathbf{r}' \rangle = \delta^3(\mathbf{r} - \mathbf{r}') \quad \langle \mathbf{p} | \mathbf{p}' \rangle = \delta^3(\mathbf{p} - \mathbf{p}') \quad (1.155)$$

The identity operator is given by:

$$\hat{I} = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r}| d^3r = \int_{-\infty}^{+\infty} |\mathbf{p}\rangle \langle \mathbf{p}| d^3p \quad (1.156)$$

The expansion of a ket is given by:

$$|\mathcal{S}\rangle = \int_{-\infty}^{+\infty} |\mathbf{r}\rangle \langle \mathbf{r} | \mathcal{S} \rangle d^3r = \int_{-\infty}^{+\infty} |\mathbf{p}\rangle \langle \mathbf{p} | \mathcal{S} \rangle d^3p \quad (1.157)$$

where $\psi(\mathbf{r}) = \langle \mathbf{r} | \mathcal{S} \rangle$ and $\phi(\mathbf{p}) = \langle \mathbf{p} | \mathcal{S} \rangle$ are the position and momentum space wave functions respectively.

The operators in each other basis given by:

$$\langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{r}}^2 \langle \mathbf{r} | \quad \langle \mathbf{p} | \hat{\mathbf{r}} = i\hbar \nabla_{\mathbf{p}}^2 \langle \mathbf{p} | \quad (1.158)$$

and their transformation function is therefore:

$$\langle \mathbf{r} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p} \cdot \mathbf{r} / \hbar} \quad (1.159)$$

1.4 Schrödinger Equation

1.4.1 Time Evolution

We consider a state represented by the state ket $|\mathcal{S}(t_0)\rangle$ at some time t_0 . At some later time $t > t_0$, we denote the state by $|\mathcal{S}, t_0; t\rangle$. At $t \rightarrow t_0$, we just write $|\mathcal{S}, t_0\rangle$. We wish to analyze the time evolution of this state, and hence we define the *time-evolution operator*.

$$\mathcal{U}(t, t_0) |\mathcal{S}, t_0\rangle = |\mathcal{S}, t_0; t\rangle \quad (1.160)$$

We demand the following properties from the time-evolution operator.

- Probability must be conserved, and hence it must be unitary.

$$\mathcal{U}(t, t_0) \mathcal{U}^\dagger(t, t_0) = \hat{I} \quad (1.161)$$

- The composition property, which states that a time evolution of from t_0 to t_2 can be achieved using two evolutions, one from t_0 to t_1 and from t_1 to t_2 , where $t_0 < t_1 < t_2$.

$$\mathcal{U}(t_2, t_0) = \mathcal{U}(t_2, t_1) \mathcal{U}(t_1, t_0) \quad (1.162)$$

We also consider an infinitesimal time-evolution operator, similar to our translation operator for position.

$$\mathcal{U}(t_0 + dt, t_0)|\mathcal{S}, t_0\rangle = |\mathcal{S}, t_0 + dt\rangle \quad (1.163)$$

And it also has properties similar to the infinitesimal translation operator.

$$\lim_{dt \rightarrow 0} \mathcal{U}(t_0 + dt, t_0) = \hat{I} \quad (1.164)$$

$$\mathcal{U}(t_0 + dt, t_0) = 1 - i\hat{\Omega} \cdot dt \quad (1.165)$$

where $\hat{\Omega}$ is a hermitian operator with dimensions of frequency, and this form satisfies all the properties we've set for the operator. From the Planck-Einstein relation, we know that $E = \hbar\omega$ and the energy is the same as the Hamiltonian, so we write the following:

$$\hat{\Omega} = \frac{\hat{H}}{\hbar} \quad (1.166)$$

and therefore the infinitesimal time-evolution operator is as follows:

$$\mathcal{U}(t_0 + dt, t_0) = 1 - \frac{i\hat{H}}{\hbar} dt \quad (1.167)$$

1.4.2 Schrödinger Equation

Postulate 1.4: Schrödinger Equation

First, we use the composition property of the time-evolution operator.

$$\mathcal{U}(t + dt, t_0) = \mathcal{U}(t + dt, t)\mathcal{U}(t, t_0) = \left(1 - \frac{i\hat{H}}{\hbar} dt\right) \mathcal{U}(t, t_0) \quad (1.168)$$

$$\frac{\mathcal{U}(t + dt, t_0) - \mathcal{U}(t, t_0)}{dt} = -\frac{i}{\hbar} \hat{H} \mathcal{U}(t, t_0) \quad (1.169)$$

We identify that the left hand side can be written as a partial derivative with respect to time.

$$i\hbar\partial_t \mathcal{U}(t, t_0) = \hat{H} \mathcal{U}(t, t_0) \quad (1.170)$$

This is the Schrödinger Equation in terms of the time-evolution operator, and for a state ket $|\mathcal{S}, t_0\rangle$, we obtain the following:

$$i\hbar\partial_t \mathcal{U}(t, t_0)|\mathcal{S}, t_0\rangle = \hat{H} \mathcal{U}(t, t_0)|\mathcal{S}, t_0\rangle \quad (1.171)$$

$$\therefore i\hbar\partial_t |\mathcal{S}, t_0; t\rangle = \hat{H} |\mathcal{S}, t_0; t\rangle \quad (1.172)$$

The Schrödinger Equation is a *postulate*, and any derivation of it must involve an assumption, here we assume the hermitian operator $\hat{\Omega} = \hat{H}/\hbar$ in the infinitesimal time-evolution operator. Let us first find solutions to the Schrödinger Equation for the time evolution operator. We consider three cases.

Case 1 — Hamiltonian is time independent, then the time evolution operator is given by:

$$\mathcal{U}(t, t_0) = \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] \quad (1.173)$$

This can be derived by performing N infinitesimal operations with $N \rightarrow \infty$ to obtain a finite time evolution.

Case 2 — Hamiltonian is time-dependent, but commutes at different times. The solution in this case is:

$$\mathcal{U}(t, t_0) = \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t) dt \right] \quad (1.174)$$

Case 3 — Hamiltonian is time-dependent and does *not commute* at different times. The formal solution in such a case is the *Dyson Series*:

$$\mathcal{U}(t, t_0) = 1 + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \int_{t_0}^{t_2} dt_3 \cdots \int_{t_0}^{t_{n-1}} \hat{H}(t_1) \hat{H}(t_2) \cdots \hat{H}(t_n) dt_n \quad (1.175)$$

1.4.3 Energy Eigenkets

We consider an observable that is compatible with the Hamiltonian, which then have simultaneous eigenkets called energy eigenkets, for which we reserve the symbol $|n\rangle$.

$$\hat{H}|n\rangle = E_n|n\rangle \quad (1.176)$$

We consider an time independent Hamiltonian, and then we can expand the time-evolution operator as follows:

$$\exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] = \sum_{n,m=1} |n\rangle \langle n| \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] |m\rangle \langle m| \quad (1.177)$$

$$= \sum_{n=1} |n\rangle \langle n| \exp \left[-\frac{i}{\hbar} E_n(t - t_0) \right] \quad (1.178)$$

And therefore the time evolution of a ket would be:

$$|\mathcal{S}, t_0; t\rangle = \mathcal{U}(t, t_0)|\mathcal{S}, t_0\rangle = \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] |\mathcal{S}, t_0\rangle \quad (1.179)$$

$$= \sum_{n=1} |n\rangle \langle n|\mathcal{S}\rangle \exp \left[-\frac{i}{\hbar} E_n(t - t_0) \right] \quad (1.180)$$

We therefore see that the coefficients evolve with time by a relative phase change, while their magnitude is constant. Now we consider the expectation value of an observable B , that need not be compatible with H .

$$\langle B \rangle = \sum_{n=1} \langle n|\mathcal{U}^\dagger(t, t_0) \hat{B} \mathcal{U}(t, t_0)|n\rangle \quad (1.181)$$

$$= \sum_{n=1} \langle n| \exp \left[\frac{i}{\hbar} \hat{H}(t - t_0) \right] \hat{B} \exp \left[-\frac{i}{\hbar} \hat{H}(t - t_0) \right] |n\rangle \quad (1.182)$$

$$= \sum_{n=1} \langle n|\hat{B}|n\rangle = \langle \mathcal{S}|\hat{B}|\mathcal{S}\rangle \quad (1.183)$$

We see that the expectation value of the observable B is independent of time, and for this reason the energy eigenket is also referred to as a *stationary state*. Similarly, for a non-stationary state, we get the expectation value to be:

$$\langle B \rangle = \sum_{n,m=1} c_n^* c_m \langle A_n | \hat{B} | A_m \rangle \exp \left[-\frac{i}{\hbar} (E_m - E_n)(t - t_0) \right] \quad (1.184)$$

1.4.4 Schrödinger's Wave Equation

We consider the position basis wave function, now with an additional time dependent factor.

$$\Psi(\mathbf{r}, t) = \langle \mathbf{r} | \mathcal{S}, t_0; t \rangle \quad (1.185)$$

The Hamiltonian operator is taken to be:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{r}, t) \quad (1.186)$$

Now, from Schrödinger's Equation we write the following:

$$i\hbar \partial_t |\mathcal{S}, t_0; t\rangle = \hat{H} |\mathcal{S}, t_0; t\rangle \implies i\hbar \partial_t \langle \mathbf{r} | \mathcal{S}, t_0; t \rangle = \langle \mathbf{r} | \hat{H} | \mathcal{S}, t_0; t \rangle \quad (1.187)$$

We know that $\langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla \langle \mathbf{r} |$, and we write the following wave equation:

$$i\hbar \partial_t \Psi(\mathbf{r}, t) = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\mathbf{r}, t) + V(\mathbf{r}, t) \Psi(\mathbf{r}, t) \quad (1.188)$$

This is fundamental equation of what is known as *wave mechanics*, called the Time-Dependent Schrödinger Wave Equation. We can further reduce this by considering a time-independent Hamiltonian, which allows us to use:

$$\langle \mathbf{r} | \mathcal{S}, t_0; t \rangle = \langle \mathbf{r} | \mathcal{S}, t_0 \rangle e^{-iE_n t/\hbar} \iff \Psi(\mathbf{r}, t) = \psi(\mathbf{r}) e^{-iE_n t/\hbar} \quad (1.189)$$

Substitute this into the Time-Dependent Wave equation, and we obtain the following equation:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \psi(\mathbf{r}) = E \psi(\mathbf{r}) \quad (1.190)$$

This is the Time-Independent Schrödinger Wave Equation, which laid the ground work for Quantum Mechanics.

1.4.5 Interpretation of the Wave Function

The wave function is regarded to be *probability density*, which follows from the fact that it is the expansion coefficient of a state ket that determine the probability of the state ket collapsing to the corresponding eigenket.

$$\rho(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2 \quad (1.191)$$

From the time-dependent wave equation, we can do the following:

$$i\hbar\Psi^*(\mathbf{r},t)\partial_t\Psi(\mathbf{r},t) = -\frac{\hbar^2}{2m}\Psi^*(\mathbf{r},t)\nabla^2\Psi(\mathbf{r},t) + V(\mathbf{r},t)|\Psi(\mathbf{r},t)|^2 \quad (1.192)$$

$$i\hbar\Psi(\mathbf{r},t)\partial_t\Psi^*(\mathbf{r},t) = \frac{\hbar^2}{2m}\Psi(\mathbf{r},t)\nabla^2\Psi^*(\mathbf{r},t) - V(\mathbf{r},t)|\Psi(\mathbf{r},t)|^2 \quad (1.193)$$

Adding the two equations (1.195) and (1.196), we get:

$$\partial_t \underbrace{(\Psi^*(\mathbf{r},t)\Psi(\mathbf{r},t))}_{|\Psi(\mathbf{r},t)|^2} = -\nabla \cdot \left(\frac{i\hbar}{2m}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) \right) \quad (1.194)$$

If we define the *probability flux* as following:

$$\mathbf{j}(\mathbf{r},t) = \frac{i\hbar}{2m}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) \quad (1.195)$$

We obtain the continuity equation for the probability flow.

$$\partial_t\rho + \nabla \cdot \mathbf{j} = 0 \quad (1.196)$$

We also find a relation between the momentum and the probability flux, which are related by the following equation:

$$\langle p \rangle = m \int_{-\infty}^{+\infty} \mathbf{j}(\mathbf{r},t) d^3r \quad (1.197)$$

2 ONE DIMENSIONAL PROBLEMS

We start by an analysis of several one dimensional, time-independent potentials. We will have a discussion of discrete and continuous energy spectrums, which is dependent on the potential analyzed, and a subsection on scattering experiments. Lastly, we introduce the harmonic oscillator potential and its quirks.

2.0.1 Bound and Scattering States

In our discussion of various potentials, we will encounter two types of solutions to the Schrödinger Equation — those with discretized energy states ; those with continuous energy spectra. These are called *bound states* and *scattering states* respectively.

The bound state solution is characterized by discrete spectrum of energy, and physically realisable solution. The energies of a bound state solution must be less than the potential energy at the extremes, i.e.

$$E < \lim_{x \rightarrow \pm\infty} V(x) \quad (2.1)$$

Scattering states are typically not normalizable solutions, and will have a continuous energy spectra. These are called as such because of the travelling wave solution one obtains when $V(x) = 0$ can be treated as scattering experiments. The energy spectra of this state must be greater than the potential energy at infinities.

$$E > \lim_{x \rightarrow \pm\infty} V(x) \quad (2.2)$$

2.0.2 Parity Operator

We introduce an operator called the parity operator, $\hat{\Pi}$. It has the effect of spatial inversion i.e.

$$\hat{\Pi}|\mathbf{r}\rangle = |-\mathbf{r}\rangle \quad (2.3)$$

It also has the effect of direction inversion for a momentum ket.

$$\hat{\Pi}|\mathbf{p}\rangle = |-\mathbf{p}\rangle \quad (2.4)$$

It is an involutory, hermitian and unitary operator, therefore:

$$\hat{\Pi} = \hat{\Pi}^\dagger = \hat{\Pi}^{-1} \quad (2.5)$$

Consider the eigenvalue equation of the operator.

$$\hat{\Pi}|\alpha\rangle = \lambda|\alpha\rangle \implies |\alpha\rangle = \lambda\hat{\Pi}|\alpha\rangle \quad (\because \hat{\Pi}^2 = \hat{I}) \quad (2.6)$$

From this, we can conclude that the eigenvalues of the parity operator are ± 1 . The eigenkets of the parity operator are called states of definite parity, or simply parity states.

There are both odd parity states $|\alpha_{-}\rangle$ with the eigenvalue -1 and even parity states $|\alpha_{+}\rangle$ with the eigenvalue 1 .

$$\text{Odd Parity} \implies \hat{\Pi}|\alpha_{-}\rangle = -|\alpha_{-}\rangle \quad (2.7)$$

$$\text{Even Parity} \implies \hat{\Pi}|\alpha_{+}\rangle = +|\alpha_{+}\rangle \quad (2.8)$$

Any arbitrary state can be written as a superposition of parity states, which we prove by considering the state $\frac{1}{2}(\hat{I} \pm \hat{\Pi})|\alpha\rangle$. Applying the parity operator would give us:

$$\hat{\Pi} \left(\frac{1}{2}(\hat{I} \pm \hat{\Pi})|\alpha\rangle \right) = \frac{1}{2}(\hat{\Pi}|\alpha\rangle \pm |\alpha\rangle) = \pm \left(\frac{1}{2}(\hat{I} \pm \hat{\Pi})|\alpha\rangle \right) \quad (2.9)$$

And therefore we can write the following:

$$|\alpha\rangle = |\alpha_{-}\rangle + |\alpha_{+}\rangle \quad (2.10)$$

$$\text{where } |\alpha_{-}\rangle = \frac{1}{2}(\hat{I} - \hat{\Pi})|\alpha\rangle \quad (2.11)$$

$$|\alpha_{+}\rangle = \frac{1}{2}(\hat{I} + \hat{\Pi})|\alpha\rangle \quad (2.12)$$

The parity operator being unitary gives us a unitary transform, we consider how it handles the position and momentum operators.

$$\langle \mathbf{r}' | \hat{\Pi} \hat{\mathbf{r}} \hat{\Pi} | \mathbf{r} \rangle = \langle -\mathbf{r}' | \hat{\mathbf{r}} | -\mathbf{r} \rangle = -\mathbf{r} \delta^3(\mathbf{r} - \mathbf{r}') = \langle \mathbf{r}' | (-\hat{\mathbf{r}}) | \mathbf{r} \rangle \quad (2.13)$$

$$\therefore \hat{\Pi} \hat{\mathbf{r}} \hat{\Pi} = -\hat{\mathbf{r}} \implies \hat{\Pi} \hat{\mathbf{r}} = -\hat{\mathbf{r}} \hat{\Pi} \quad (2.14)$$

Similarly, we get $\hat{\Pi} \hat{\mathbf{p}} \hat{\Pi} = -\hat{\mathbf{p}}$ and $\hat{\Pi} \hat{\mathbf{p}} = -\hat{\mathbf{p}} \hat{\Pi}$. Now, we consider the commutability of the parity and Hamiltonian operators.

$$[\hat{\Pi}, \hat{H}] = \hat{\Pi} \hat{H} - \hat{H} \hat{\Pi} = \hat{\Pi} \left(\frac{1}{2m} \hat{\mathbf{p}}^2 + V(\mathbf{r}) \right) - \left(\frac{1}{2m} \hat{\mathbf{p}}^2 + V(\mathbf{r}) \right) \hat{\Pi} \quad (2.15)$$

$$= \left(\frac{1}{2m} (-\hat{\mathbf{p}})^2 + V(-\mathbf{r}) \right) \hat{\Pi} - \left(\frac{1}{2m} \hat{\mathbf{p}}^2 + V(\mathbf{r}) \right) \hat{\Pi} \quad (2.16)$$

$$= (V(-\mathbf{r}) - V(\mathbf{r})) \hat{\Pi} \quad (2.17)$$

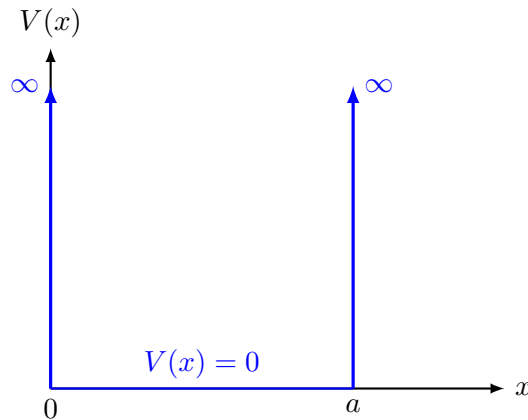
We see that the parity and Hamiltonian operators commute when the potential is symmetric/even i.e. $V(\mathbf{r}) = V(-\mathbf{r})$. This means that we have simultaneous eigenkets of energy and parity, which gives a very useful tool for even potentials.

$$\hat{\Pi}|n\rangle = \pm|n\rangle \implies \langle x | \hat{\Pi} | n \rangle = \pm \langle x | n \rangle \implies \langle -x | n \rangle = \pm \langle x | n \rangle \quad (2.18)$$

2.1 Infinite Square Well

We start with a very basic potential: a particle confined in a box, which is represented by the infinite square well potential.

$$V(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 < x < a \\ \infty, & x > a \end{cases} \quad (2.19)$$



We see that $E > V(\pm\infty)$, and therefore it is a bound state. Let us begin by constructing the Hamiltonian operator for this system.

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V} \xrightarrow{\hat{V}=0} \frac{\hat{p}^2}{2m} \quad (2.20)$$

Now, we look for energy eigenkets of the system to find the energy spectrum.

$$\hat{H}|n\rangle = E_n|n\rangle \implies \frac{1}{2m}\hat{p}^2|n\rangle = E_n|n\rangle \implies \langle x|\hat{p}^2|n\rangle = 2mE_n\langle x|n\rangle \quad (2.21)$$

We use our result from Equation (1.143) to write $\hat{p}^2$ in position space.

$$\partial_x^2\langle x|n\rangle = -\frac{2mE_n}{\hbar^2}\langle x|n\rangle \quad (2.22)$$

If we let $k = \sqrt{2mE_n}/\hbar$ (we will use this substitution a lot, so unless mentioned, k refers to this!), we solve the PDE to get:

$$\langle x|n\rangle = A\sin(kx) + B\cos(kx) \quad (0 < x < a) \quad (2.23)$$

The infinitely tall potential walls at $x = 0$ and $x = a$ dictate that the particle stays in the region $0 < x < a$, which gives us the boundary conditions that $\langle 0|n\rangle = \langle a|n\rangle = 0$. Applying the boundary conditions gives us:

$$A\sin(ka) + B\cos(ka) = 0 \quad ; \quad B = 0 \quad (2.24)$$

This implies that ka take integral values of π to satisfy the boundary conditions.

$$ka = \frac{a\sqrt{2mE_n}}{\hbar} = n\pi \quad (2.25)$$

This finally gives us the energy spectrum of the particle in a box.

$$\therefore E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} = n^2 E_1 \quad (2.26)$$

Therefore we should get $\langle x|n\rangle$ as following.

$$\langle x|n\rangle = A \sin\left(\frac{n\pi}{a}x\right) \quad (0 < x < a) \quad (2.27)$$

Now, to determine the constant A , we use the fact that $|n\rangle$ must be normalized, therefore:

$$1 = \langle n|n\rangle = \int_{-\infty}^{+\infty} \langle n|x\rangle \langle x|n\rangle dx \quad (2.28)$$

$$= \int_0^a A^2 \sin^2\left(\frac{n\pi}{a}x\right) dx = \frac{A^2}{2}a \quad (2.29)$$

$$\therefore A = \sqrt{\frac{2}{a}} \quad (2.30)$$

Putting this back where it belongs, we finally get that:

$$\langle x|n\rangle = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) \quad (0 < x < a) \quad (2.31)$$

And therefore, the energy eigenkets of the infinite square well are:

$$\therefore |n\rangle = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right) |x\rangle dx \quad (2.32)$$

Now, let's tackle the problem of time evolution of the state ket in this system. Since we have a time-independent Hamiltonian, we can write the time-evolution operator as:

$$\mathcal{U}(t) = e^{-i\hat{H}t/\hbar} \quad (2.33)$$

Therefore the ket expansion can be written as:

$$|\mathcal{S}(t)\rangle = \mathcal{U}(t)|\mathcal{S}\rangle = \sum_{n=1} |n\rangle \langle n|e^{-i\hat{H}t/\hbar}|\mathcal{S}\rangle \quad (2.34)$$

Now, from Equation (1.178) we can write further:

$$|\mathcal{S}(t)\rangle = \sum_{n=1} c_n e^{-iE_n t/\hbar} |n\rangle \quad (2.35)$$

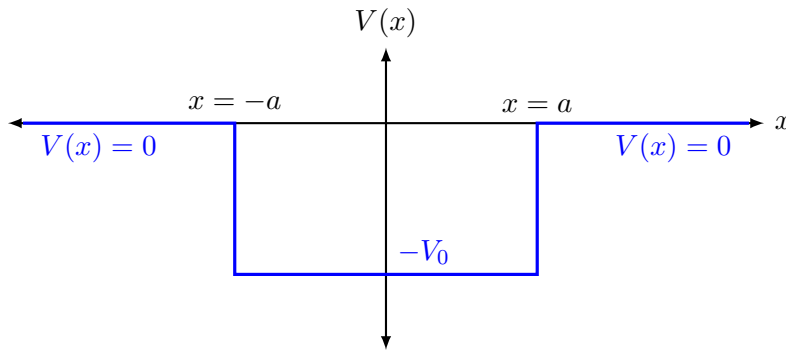
where $c_n = \langle n|\mathcal{S} \rangle$ are the expansion coefficients, given by:

$$c_n = \langle n|\mathcal{S} \rangle = \int_{-\infty}^{+\infty} \langle n|x \rangle \langle x|\mathcal{S} \rangle dx = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right) \langle x|\mathcal{S} \rangle dx \quad (2.36)$$

2.2 Finite Square Well

We once again consider a potential well, but this time the walls of the well are not infinitely tall, which allows the position to take values that would be restricted in the infinite square well.

$$V(x) = \begin{cases} 0, & |x| > a \\ -V_0, & |x| < a \end{cases} \quad V_0, a > 0 \quad (2.37)$$



Upon further inspection, we notice that this potential has both a bound state and a scattering state. We shall first consider the bound state with energy $E < 0$.

2.2.1 Bound State

We once again begin by constructing the Hamiltonian and trying to find the energy eigenkets. In this case, we have two different values of potential, and therefore two different values of the Hamiltonian: $\hat{H} = \hat{p}^2/2m$ in the region $|x| < a$ and outside it, we have the $\hat{H} = \hat{p}^2/2m - V_0$. Lets tackle the region $0 < x < a$ first.

$$\hat{p}^2|n\rangle = 2mE_n|n\rangle \implies \partial_x^2 \langle x|n\rangle = -k^2 \langle x|n\rangle \quad (2.38)$$

Here, $k^2 = 2mE_n/\hbar^2$. The solution to this would be:

$$\langle x|n\rangle = A \sin kx + B \cos kx \quad (0 < x < a) \quad (2.39)$$

Now, for the region $x > a$ we have:

$$\left(\frac{1}{2m}\hat{p}^2 - V_0\right)|n\rangle = E_n|n\rangle \implies \partial_x^2 \langle x|n\rangle = -l^2 \langle x|n\rangle \quad (2.40)$$

Here, $l^2 = 2m(E_n + V_0)/\hbar^2$ and the solution to this would be:

$$\langle x|n\rangle = Ce^{lx} + De^{-lx} \quad (x > a) \quad (2.41)$$

Now, if you have noticed, we are dealing with a symmetric potential and therefore the energy eigenkets are also eigenkets of the parity operator. So we have even or odd $\langle x|n\rangle$ functions.

Even Parity State

For even parity states, we have the following relations:

$$\langle x|n_+\rangle = \langle -x|n_+\rangle \quad (2.42)$$

$$x < -a \implies \langle x|n_+\rangle = Ae^{-kx} + Be^{kx} \implies A = B \quad (2.43)$$

$$-a < x < 0 \implies \langle x|n_+\rangle = -C \sin(lx) + D \cos(lx) \implies C = 0 \quad (2.44)$$

We also shave off the e^{-kx} term for $x > a$ and e^{kx} term for $x < -a$ to keep our kets normalizable. Therefore:

$$\langle x|n_+\rangle = \begin{cases} Ae^{kx}, & x < -a \\ D \cos(lx), & |x| < a \\ Ae^{-kx}, & x > a \end{cases} \quad (2.45)$$

We also have to preserve the continuity of $\langle x|n_+\rangle$ and its derivative, so we have the following relations:

$$Ae^{ka} = D \cos(la) \quad ; \quad -Ake^{ka} = -Dl \sin(la) \quad (2.46)$$

From which we get the following relation between k and l .

$$l \tan(la) = k \quad (2.47)$$

Now, using the definitions of l and k , we write that:

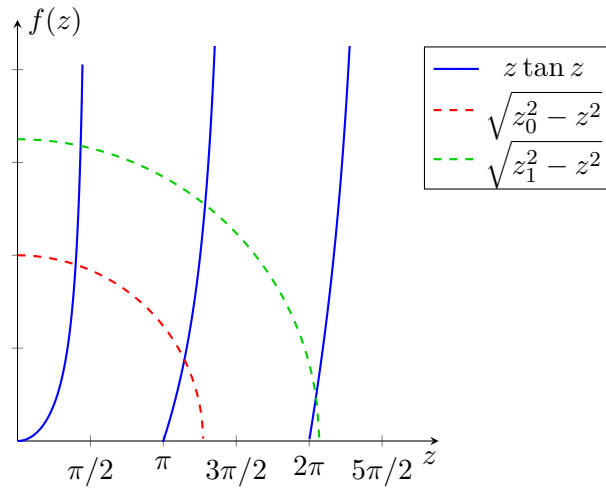
$$l^2 + k^2 = \frac{2mV_0}{\hbar^2} \quad (2.48)$$

Using this, we write:

$$(la) \tan(la) = \sqrt{\left(\frac{\sqrt{2mV_0}a}{\hbar}\right)^2 - (la)^2} \quad (2.49)$$

Let $z = la$ and $z_0^2 = 2mV_0a/\hbar^2$, then we get the following transcendental equation, whose solutions we obtain by numerical methods or drawing plots.

$$z \tan z = \sqrt{z_0^2 - z^2} \quad (2.50)$$



We see that for higher values of z_0 , we have more solutions to the even bound state. As $z_0 \rightarrow \infty$, we would see intersections at exactly odd multiples of $\pi/2$, which would give us the energy:

$$E_n + V_0 = \frac{n^2 \hbar^2 \pi^2}{8ma^2}, \quad n = 1, 3, 5, \dots \quad (2.51)$$

Odd Parity State

A similar analysis of the odd parity state would give us:

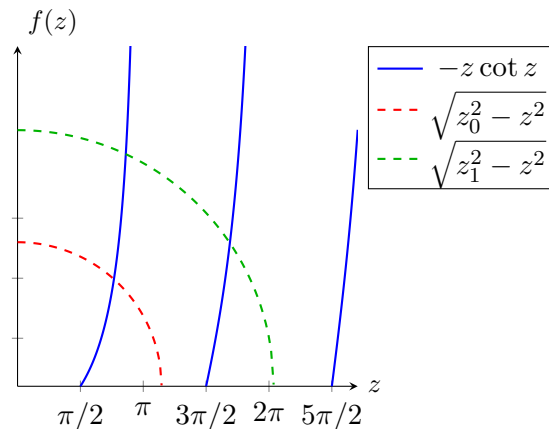
$$\langle x | n_- \rangle = \begin{cases} -Ae^{kx}, & x < -a \\ C \sin(lx), & |x| < a \\ Ae^{-kx}, & x > a \end{cases} \quad (2.52)$$

And the continuity condition gives the following equation:

$$-l \cot(la) = k \quad (2.53)$$

Which we rewrite by taking the same z and z_0 ,

$$-z \cot z = \sqrt{z_0^2 - z^2} \quad (2.54)$$



Observe that for $z_0 < \pi/2$, we would have no solutions to the odd bound state. There may always be an even bound state, but there may not be an odd one. And if we have $z_0 \rightarrow \infty$, we see the intersections at exactly the multiples of π , or we can say the even multiples of $\pi/2$. Therefore, the energy in this state would be:

$$E_n + V_0 = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 2, 4, 6, \dots \quad (2.55)$$

We see the energies distributed evenly between the odd and even states, with the even state getting odd multiples of n and vice-versa.

If you remember, we can write any ket as a sum of an odd and even parity ket, and we do this with the energy eigenkets. So we write:

$$|n\rangle = |n_+\rangle + |n_-\rangle \quad (2.56)$$

With the Hamiltonian being time-independent, the time-evolution operator is $e^{-i\hat{H}t/\hbar}$ and the state ket expansion of the system would be:

$$|\mathcal{S}(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |n\rangle \quad (2.57)$$

$$\text{where } c_n = \langle n | \mathcal{S} \rangle \quad (2.58)$$